DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 11

# ✈ Chapter 11: Radar Principles

Radio Detection And Ranging

## 📋 Contents

1. Introduction
2. Types of Radar
3. Radar Frequencies
4. Pulse Technique (PRI, PRF)
5. Range Measurement (Echo Principle)
6. Maximum & Minimum Range
7. Factors Affecting Range
8. Radar Resolution
9. Moving Target Indication (MTI)
10. Radar Antennae
11. Practice Questions (14 Q)

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## 1. Introduction

Radar = **RA**dio **D**etection **A**nd **R**anging. Developed before WWII, used both for military and civil aviation. Today radar is essential for ATC, weather warning (AWR), surface movement (ASMI), and navigation (DME is secondary radar).

### 📡 Radar Applications in Aviation

| Use | System |
|---|---|
| ATC surveillance | ASR, TAR, PAR, ASMI |

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| | |
|---|---|
| Aircraft identification + altitude | SSR (Secondary Surveillance Radar) |
| Range measurement | DME (secondary radar) |
| Weather depiction | AWR (Airborne Weather Radar) |
| Weather info to ATC | Ground weather radars |

## 2. Types of Radar

### Primary vs Secondary Radar

| Type | How it Works | Examples |
|---|---|---|
| **Primary Radar** | Transmits pulse, detects reflected echo from target. **One frequency** . Does not require target cooperation. | ATC surveillance, AWR, PAR, ASMI |
| **Secondary Radar** | Transmits interrogation pulse; target (transponder) **transmits its own reply** on different frequency. Requires transponder. | SSR, DME |

## 3. Radar Frequencies

### Why VHF and Above?

- Free from external noise/static and ionospheric scatter
- Shorter wavelengths produce **narrow, efficient beams** for target discrimination and bearing
- Shorter wavelengths allow shorter pulses (better range resolution)
- Efficient reflection: shorter λ reflected more efficiently from objects

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## 4. Pulse Technique

The duration of the pulse is equal to the **pulse length or width**. Although a pulse is of short width (time) it can contain many cycles.

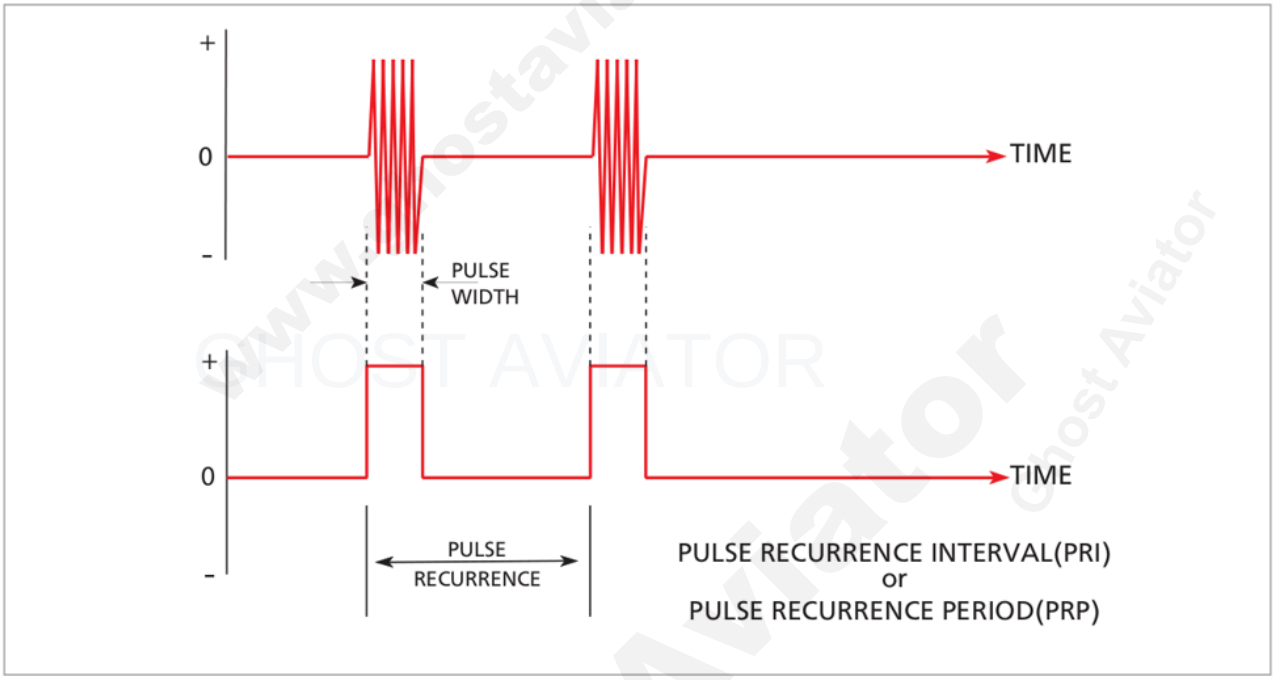



*Figure 11.2 Pulse technique*

**Pulse Recurrence Interval (PRI)** is the **time interval** between two pulses.

**Pulse Recurrence Frequency (PRF)** is the number of pulses transmitted in one second **(pps).**

**Example.** If the PRF is 250 pps what is the PRI of the transmission?

PRI = 1 / 250 s

PRI = 1 000 000 / 250 µs = 4000 µs

*Fig 11.2: Pulse technique — pulse width, PRI, PRF illustrated*

### Key Pulse Parameters

| Term | Definition | Formula |
|---|---|---|
| **Pulse Width (PW)** | Duration of each transmitted pulse | — |
| **PRF / PRR** | Pulse Recurrence Frequency — pulses per second (pps) | — |
| **PRI / PRP** | Pulse Recurrence Interval/Period — time between pulses | PRI = 1/PRF |

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### ⚡ Example Calculation

PRF = 250 pps → PRI = 1/250 s = 1,000,000/250 µs = **4000 µs**

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## 5. Range Measurement (Echo Principle)

### Distance Measurement - Echo Principle

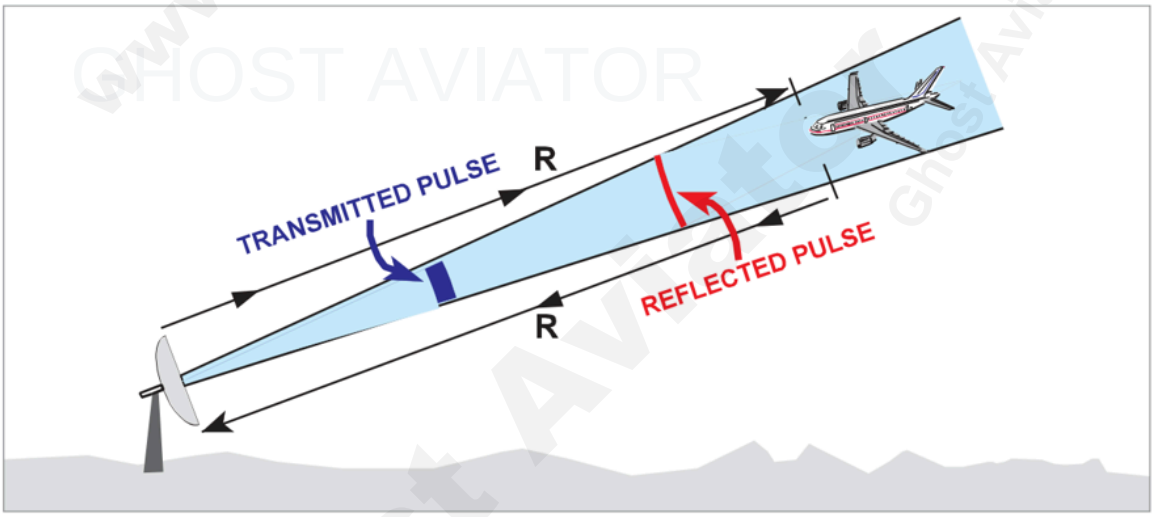



*Figure 11.3*

The distance to an object is found by timing the interval between the instant of the pulse's transmission and its return as an echo; this is shown in *Figure 11.3*.

For example, if the echo (the time between transmission and reception) is 500 µs then:

*Fig 11.3: Echo principle — timing the round-trip of the pulse determines range*

### 📡 Range Calculation

**Range (NM) = Echo time (µs) / 12.36**

**Range (km) = Echo time (µs) × 300 / 2**

One radar mile (1 NM out and back) = **12.36 µs**

Speed of radio waves: c = 3 × 10&sup8; m/s = 162,000 NM/s

### ⚡ Worked Example

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Echo time = 500 µs:

Range = 500/12.36 = **40.5 NM** | Range = 500 × 300/2 = **75 km**

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## 6. Maximum & Minimum Range

### ⚠ Maximum Range — Limited by PRF

Each pulse must return from the most distant target BEFORE the next pulse is transmitted.

**Max Range (NM) = 81,000 / PRF(pps)  [or: PRF = 81,000 / Max_Range_NM]**

Max Range (km) = 150,000 / PRF(pps)

**Rule:**

Greater range required → lower PRF.

### ⚠ Minimum Range — Limited by Pulse Width

Target must be far enough that its echo returns AFTER the pulse tail leaves the transmitter.

In 1 µs, radio waves travel 300 m (150 m each way).

**Min Range (m) = Pulse Width (µs) × 150**

Short pulses needed for short range; long pulses for long range.

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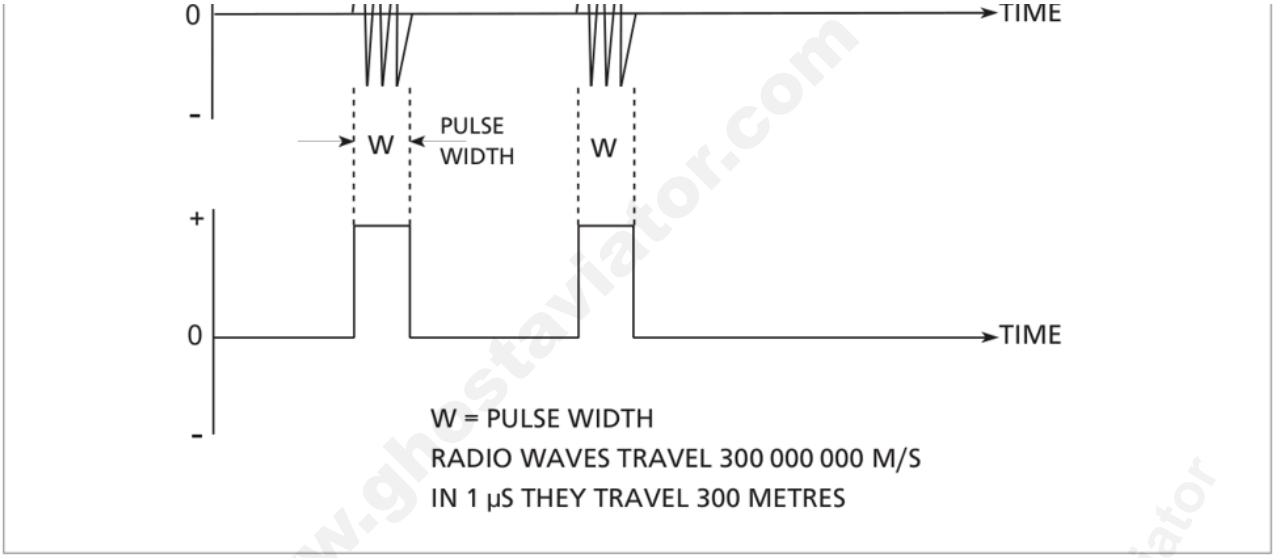



*Figure 11.7 Pulse Width Decides Minimum Range*

- **Restoration Time** is a design factor that affects the time taken for a receiver to recover to normal after transmission has occurred.

*Fig 11.7: Pulse width determines minimum range — target must be beyond 150m per µs of pulse width*

### ⚡ PRF Examples

| Requirement | Calculation | Answer |
|---|---|---|
| Max range 187 km, find PRF | PRF = 300,000,000/(2×187,000) | **802 pps** |
| Max range 200 NM, find PRF | PRF = 81,000/200 | **405 pps** |
| PRF = 400 pps, find max range NM | 81,000/400 | **203 NM** |
| PRI = 2100 µs, find max range NM | PRF=1/2100µs=476pps; 81,000/476 | **170 NM** |

## 7. Factors Affecting Range & Accuracy

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## Primary Radars

The pulses are concentrated into the beam dimensions designed for the particular radar. The beam uses the 'echo' **principle to determine range and the 'searchlight' principle to indicate bearing or height**. *Figure 11.4* shows the Plan Position Indicator (PPI) display and *Figure 11.5* shows the ATC radar antennae. The long structures at the top of the primary radar antennae are the secondary radar antennae.

The transmitter and receiver share the same antenna. The receiver is energized to accept 'echoes' from objects in the pulses' path as soon as the transmitter pulse exits the antenna. The reflected pulses are very weak due to the double journey.

The shape and size of the radar antennae determines the size of the main and side lobes as well as the width of the radar beam generated by the system. The larger the aerial, the narrower will be the beam.



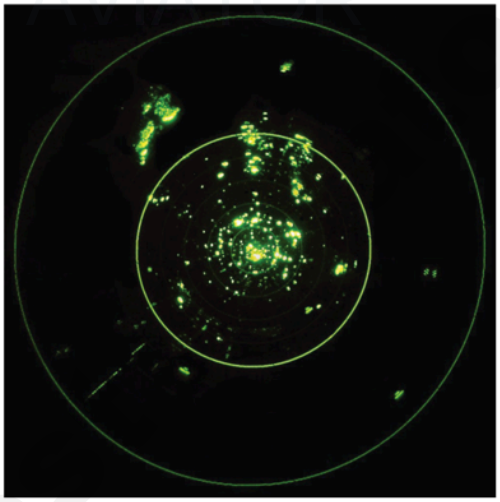


*Figure 11.4 A PPI display of primary raw radar*

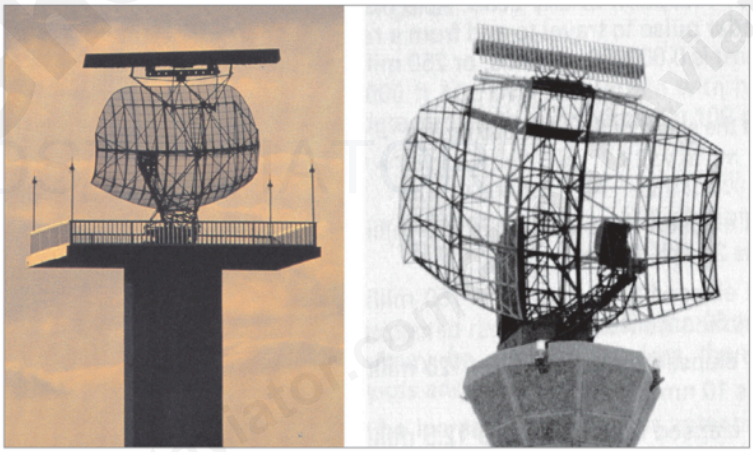


*Figure 11.5 Typical radar antennae*



*Fig 11.4: PPI display and Fig 11.5: ATC radar antennae (primary radar with SSR on top)*

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of reflected waves. The side of the fuselage has a better aspect than the nose of the aircraft.

- **Aircraft height and the height of the radar head.** Radar transmissions, because of their frequency bands, travel in straight lines and give line of sight ranges, plus a little extra due to atmospheric refraction. Thus the curvature of the earth causes much of the surface to be in shadow. Therefore, higher flying aircraft are more likely to be detected because they are above that shadow. Intervening high ground also will screen low flying aircraft from detection. The higher the radar head can be positioned, the greater that radar's range and the less effect intervening high ground will have on stopping signals and reducing its range. The following formula can be used to calculate the maximum theoretical radar range:

  Max. Theoretical range (NM) = $1.23 \times (\sqrt{H_{TX}} + \sqrt{H_{RX}})$

  $H_{TX}$ = height of radar station in feet AMSL; $H_{RX}$ = height of target in feet AMSL.

- **Wavelength and attenuation by raindrops**




*Fig 11.6: Attenuation by raindrops vs wavelength — shorter wavelengths absorbed more*

### Factors Affecting Primary Radar Range

| Factor | Effect |
|---|---|
| **Transmit power** | Power ∝ range&sup4; → to **double range**, power must increase by factor of **16** |
| **Target characteristics** | Metal > wood; larger objects > smaller; aspect angle matters |
| **Aircraft height** | LOS system — higher aircraft detected at greater range; high ground screens low-level aircraft |
| **Wavelength vs rain** | λ >10 cm: negligible attenuation; λ 4–10 cm: significant in tropical rain; λ <4 cm: significant in temperate rain |
| **Super-refraction** | Temperature inversion + humidity decrease with height → waves refract DOWN → beyond LOS range |
| **Sub-refraction** | Waves refract UP → reduced range |

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| | |
|---|---|
| **Restoration time** | Recovery time of receiver after transmission — design factor affecting minimum range |
| **Fly-back / dead time** | CRT trace return time → reduces practical range below theoretical maximum |

### ⚡ Wavelength Guide for Radar Systems

- ASMI (Surface Movement): **1.75–2 cm** (15–17 GHz, SHF)
- AWR / PAR: **3 cm** (9375 MHz, SHF)
- Aerodrome Surveillance: **3 or 10 cm**
- Terminal Surveillance: **10, 23 or 50 cm**
- Long range ASR: **10–50 cm**
- Do not use λ < 3 cm for long range systems (too much rain attenuation)

### 📡 LOS Range Formula for Radar

$$\textbf{Max Range (NM)} = 1.23 \times (\sqrt{H_{TX}} + \sqrt{H_{RX}})$$

H in feet AMSL. (Note: 1.25 used for VOR/VDF; radar uses 1.23)

## 8. Radar Resolution

### 📡 Resolution Rectangle

A point target appears as a

**rectangle**

on the PPI — stretched radially (by pulse width) and in azimuth (by beamwidth).

| Dimension | Determined by | Detail |
|---|---|---|
| Radial (range) resolution | Half pulse width | 1 μs PW → 150 m radial stretch |
| Azimuth resolution | Full beamwidth | 3° BW at 120 km → 6 km azimuth stretch (1:60) |

To improve resolution:

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**shorter pulse + narrower beam**

(larger antenna). Narrowing beam requires larger aerial.

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## 9. Moving Target Indication (MTI)

### MTI

- Eliminates returns from stationary objects (hills, buildings) — shows only moving targets (aircraft)
- Risk:
  **second trace returns**
  — echoes from beyond selected range appear as false moving targets
- Solution:
  **Jitter the PRF**
  (change PRI between consecutive pulses) → removes second trace returns

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## 10. Radar Antennae

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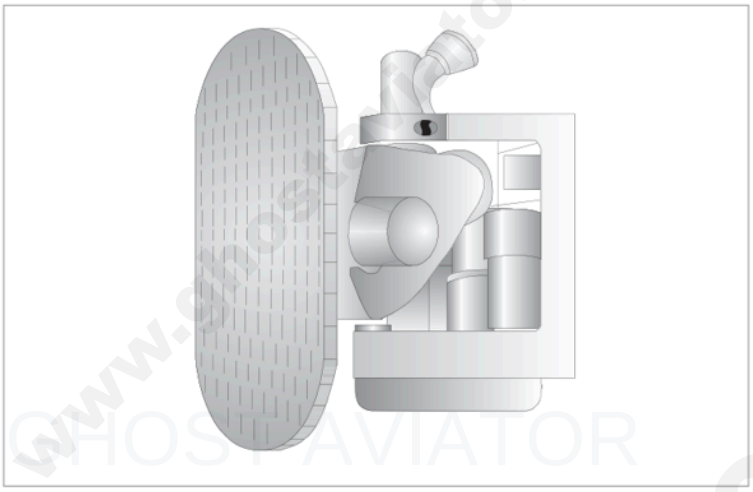


Figure 11.9 Airborne weather radar antenna

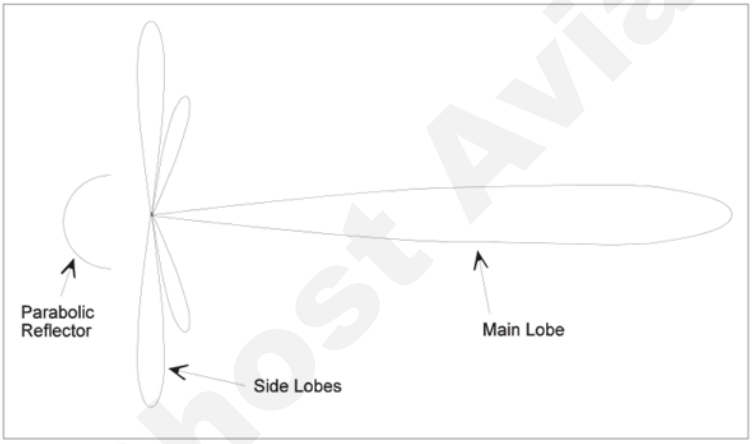



Figure 11.10 Typical radiation pattern



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*Fig 11.8: Radar antennae types | Fig 11.9: AWR flat-plate antenna | Fig 11.10: Radiation pattern with main and side lobes*

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### Antenna Types

| Type | Features | Use |
|---|---|---|
| Microwave Horn | Often used as feed for parabolic reflectors | General purpose |
| Parabolic Reflector | Dish; main lobes + side lobes; needs side-lobe suppression | ATC, AWR |
| Slotted Planar Array (Flat Plate) | Narrower beam, much smaller side lobes; less power needed; better resolution | Modern AWR, PAR |

**Larger Antenna = Narrower Beam = Better Resolution**

But a larger antenna in an aircraft is impractical → hence shorter wavelengths are used in aircraft (AWR: 3 cm).

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## 11. Practice Questions

**Q1. The factor which determines the MAXIMUM range of a radar is:**

**(a)** pulse repetition rate

**(b)** pulse width

**(c)** power

**(d)** beamwidth

**Answer: (a)**
Maximum range is determined by the **pulse repetition rate (PRF)**. The next pulse cannot be sent until the first returns from maximum range. Lower PRF = greater max range.

**Q2. The main advantage of continuous wave (CW) radars is:**

**(a)** no maximum range limitation

**(b)** better range resolution

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**(c)** no minimum range limitation

**(d)** better bearing accuracy

> **Answer: (c)**
> CW radars have **no minimum range limitation** (the radio altimeter uses CW). Pulsed radars have a minimum range determined by pulse width.

### Q3. If the PRF of a primary radar is 500 pps, the maximum range will be:

**(a)** 324 NM

**(b)** 300 NM

**(c)** 162 NM

**(d)** 600 NM

> **Answer: (c)**
> Max range = 81,000 / PRF = 81,000 / 500 = **162 NM**.

### Q4. To double the range of a primary radar requires power to be increased by a factor of:

**(a)** 2

**(b)** 4

**(c)** 8

**(d)** 16

> **Answer: (d)**
> Power ∝ range&sup4;. To double range: power increase = 2&sup4; = **16**.

### Q5. Echo time is 1720 µs. Range of target:

**(a)** 139 km

**(b)** 258 km

**(c)** 278 km

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**(d)** 516 km

> **Answer: (b)**
> Range = 1720 × 300 / 2 = **258,000 m = 258 km**.

### Q6. Max range required: 100 NM. Maximum PRF?

**(a)** 1620 pps

**(b)** 1234 pps

**(c)** 617 pps

**(d)** 810 pps

> **Answer: (d)**
> PRF = 81,000 / 100 = **810 pps**.

### Q7. PRI = 2100 µs. Maximum radar range:

**(a)** 170 NM

**(b)** 315 NM

**(c)** 340 NM

**(d)** 630 NM

> **Answer: (a)**
> PRF = 1,000,000 / 2100 = 476 pps. Max range = 81,000 / 476 ≈ **170 NM**.

### Q8. To improve resolution of a radar display:

**(a)** narrow pulse width and narrow beamwidth

**(b)** high frequency and large reflector

**(c)** wide beamwidth and wide pulse width

**(d)** low frequency and narrow pulse width

> **Answer: (a)**
> Better resolution = shorter pulse width (radial) + narrower beamwidth (azimuth). Answer:

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**narrow PW + narrow BW**.

### Q9. Advantage of phased array (slotted antenna):

**(a)** better resolution

**(b)** less power required

**(c)** reduced side lobes and clutter

**(d)** all of the above

**Answer: (d)**
Slotted planar array: narrower beam → better resolution, less power needed, reduced side lobes. All of the above.

### Q10. Echo received 900 μs after transmission. Range to target:

**(a)** 73 NM

**(b)** 270 NM

**(c)** 135 NM

**(d)** 146 NM

**Answer: (a)**
Range = 900/12.36 = **72.8 ≈ 73 NM**.

### Q11. Factor limiting MINIMUM detection range of radar:

**(a)** pulse repetition interval

**(b)** transmitter power

**(c)** pulse width

**(d)** pulse repetition frequency

**Answer: (c)**
Minimum range is limited by **pulse width**. Target must be far enough that its echo returns after the pulse has finished transmitting.

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**Q12. Doppler MTI generates second trace returns. These are removed by:**

**(a)** using different frequency for Tx/Rx

**(b)** jittering the PRF

**(c)** regular pulsewidth changes

**(d)** limiting power output

**Answer: (b)**
Second trace returns are removed by **jittering the PRF** (varying PRI between pulses). This prevents false targets from appearing within range.

**Q13. Radar max range: 12 km. Maximum PRF?**

**(a)** 25,000 pps

**(b)** 6,700 pps

**(c)** 12,500 pps

**(d)** 13,400 pps

**Answer: (c)**
PRF = 300,000,000 / (2 × 12,000) = 300,000,000 / 24,000 = **12,500 pps**.

**Q14. Bearing of primary radar measured by:**

**(a)** phase comparison

**(b)** searchlight principle

**(c)** lobe comparison

**(d)** DF techniques

**Answer: (b)**
Primary radar bearing uses the **searchlight principle** — a narrow rotating beam; direction of target = direction of beam when echo is received.

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